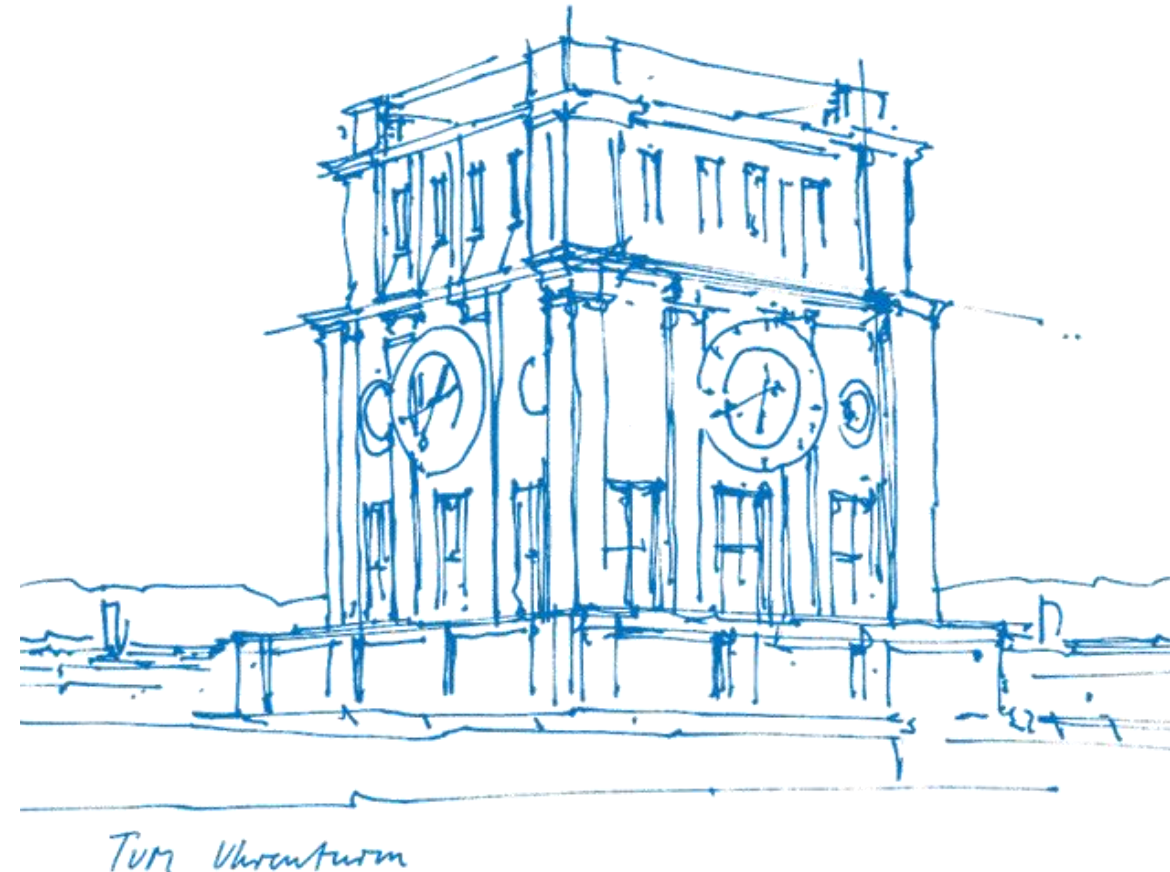


Tutorial 06

Expected Value, (Co)Variance, Correlation Coefficient, Memoryless Distributions

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About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week
 - 12:00-14:00 Tue, 03.11.018 (Group B33)
 - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code 

➤ **PART I.** Recap of the Lecture

➤ **PART II.** Exercises from the Exercise Sheet

(Don't forget to register for the exam on TUMonline!)



I. Recap

• (Review)

- ① - Expected Value *“Average value of a RV”*
- ② - Variance *“The fluctuation of a RV”*
- ③ - Covariance *“How are two RVs related”*

• Correlation Coefficient

- ④ - Definition
- (⑤ - Partial Correlation Coefficient)

• Quantile

- ⑥ - Quartile, median

6	11	24	37	49	55	60	73	89	92	104
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Quartile: Q1 = 24

Quartile: Q3 = 89

Median: M = 55

• Memoryless Distributions

- ⑦ - Discrete / Continuous memoryless distributions

II. Exercise Sheet

Exercise 1 (spurious correlation and confounding). A data scientist is investigating the relationship between daily ice cream sales (I) and the number of drowning incidents (D) in a coastal town during the summer. They collect data for 30 days and observe the following joint probabilities for simplified categories (Low/High sales, Low/High incidents):

$P(I = i, D = d)$	$D = \text{Low}$	$D = \text{High}$	$P(I = i)$
$I = \text{Low}$	0.30	0.10	0.40
$I = \text{High}$	0.15	0.45	0.60
$P(D = d)$	0.45	0.55	1.00

To simplify calculations, let's assign numerical values: Low=0, High=1 for both I and D .

- Calculate $\mathbb{E}[I]$, $\text{var}[I]$, $\mathbb{E}[D]$, $\text{var}[D]$, $\text{cov}[I, D]$, and $\rho[I, D]$.
- Based on $\rho[I, D]$, what do you conclude about the relationship between ice cream sales and drowning incidents?
- A meteorologist suggests that the average daily temperature (T) is a confounding variable that could be responsible for the observed correlation. Explain conceptually, how a third variable could “create” the observed correlation between I and D . How would you check this and what data would you need? How does that link to the famous saying that “correlation does not imply causation?”

II. Exercise Sheet

Exercise 2 (value at risk (VaR)). An investment analyst models the daily percentage return R of a particular stock as a continuous random variable with pdf

$$p_R(r) = \begin{cases} \frac{2}{15}(r+3) & \text{if } -3 \leq r < 0, \\ \frac{1}{5}(2-r) & \text{if } 0 \leq r \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Since r is in percent, $r = 1$ means a 1% return. The *Value at Risk (VaR)* at confidence level α (or, expressed in percentages, at a $(100\alpha)\%$ confidence level) is the maximum percentage loss L such that the probability of experiencing a loss greater than L is α . Mathematically, if $L > 0$ is the VaR, then $P(R \leq -L) = \alpha$.

- (i) Verify that $p_R(r)$ is a valid pdf.
- (ii) Calculate the cumulative distribution function (CDF), $F_R(r)$. Be sure to define it for all $r \in \mathbb{R}$.
- (iii) Explain how the $(100\alpha)\%$ VaR (expressed as a positive loss percentage) relates to the α -quantile, $r_\alpha = F_R^{-1}(\alpha)$, of the return distribution R .
- (iv) Determine the 5% Value at Risk (VaR) for this investment.
- (v) Calculate the median daily return, the expected daily return, and the daily return that is “most likely,” i.e., the maximum of the pdf. Explain your findings in words.
- (vi) If an investor has \$250,000 in this stock, what is the dollar amount corresponding to the 5% daily VaR you calculated in part (iv)?

II. Exercise Sheet

Exercise 3 (lack of memory of the exponential distribution). Let $X : (\Omega, \mathcal{A}, P) \rightarrow (\mathbb{R}_{>0}, \mathcal{B})$ be a random variable such that $X > 0$ with strictly positive density function $p_X(x) > 0$ for all $x \in \mathbb{R}_{>0}$. Show that X is exponentially distributed, i.e., $X \sim \text{Exp}(\lambda)$ for some $\lambda > 0$, if and only if

$$P(X > t + s \mid X > s) = P(X > t) \quad \forall s, t \geq 0. \quad (\text{A})$$

[Hint: Note that the cdf of $\text{Exp}(\lambda)$ is $F_X(x) = 1 - e^{-\lambda x}$ and that $P(X > x) = 1 - P(X \leq x) = 1 - F_X(x)$. Use that the conditional probability can be written as $P(X > t + s) / P(X > s)$. For “ $\Leftarrow$ ” consider $P(X > x) = \int_x^\infty f_X(a) da =: f(x)$ and $\varphi(s) = \frac{f(t+s)}{f(s)}$. How does φ depend on s ?]

This states that the exponential function has “no memory” in the sense that an event occurring in the future is independent of how much time has already elapsed: Given that it has not occurred up until time s , the probability of it not occurring in an additional time t is the same as if we started from scratch and waited for time t . The distribution “forgets” how much time has already elapsed. The statement also says that it is the only continuous distribution with that property (on $\mathbb{R}_{>0}$).

II. Exercise Sheet



Exercise 4 (does treatment help? coding + open ended parts).

Consider the following settings.

- (i) As data scientist in residence at the university clinic, you are sent a dataset of a clinical study containing three columns with headers *treatment* (T), *survival* (S), *precondition* (P), all containing only binary values. The email says: “Please analyze whether the treatment helped.” You can find the data on Moodle called `hospital.csv`. You can use any programming language you like and I somewhat expect you use the LLM/coding assistant/agent of your choice. Do you agree with it?
 - (a) Try to formalize the question in your own words. Take note of what choices and assumptions you have to make in your modeling.
 - (b) Compute the correlation coefficient between treatment and survival. What do you conclude?
 - (c) You are not sure what “precondition” means, but you suspect that people with a precondition may be better off. Compute the correlation coefficient between precondition and survival. What do you conclude?
 - (d) You decide to compute the expected difference in survival probability when getting treatment, compared to no treatment, separately within patients with and without preconditions. Compute these differences and provide a first answer for your task.
 - (e) The doctors reply: “But overall, among all people, does the treatment help? Are we sure?” What do you reply?
- (ii) While you draft your reply, the doctors call to let you know they made a mistake and the third column should have been named *complications during treatment* instead of *precondition*. What would you change about your analysis in steps (a) to (e) from (i)? Why?

II. Exercise Sheet



Exercise 5 (★ bonus ★). Let $(\Omega, \mathcal{A}, P)$ be a probability space and X, Y, Z real-valued random variables. By means of a counter-example, show the following:

- (i) If X and Y are partially uncorrelated given Z , they are not necessarily independent given Z .
[Hint: Look at Exercise 4.4 and $Z \perp\!\!\!\perp X$.]
- (ii) If X and Y are independent given Z , they are not necessarily partially uncorrelated given Z .
[Hint: Consider $Z \sim \mathcal{U}_{(0,1)}$, $X = Z^2$ and $Y = Z^3$.]